Learning Which Lessons to Show a CTF Agent: Learned Choice Wins on Synthetic Tasks, Not Yet on Real CTFs

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***Abstract*—LLM agents can keep a memory of short technique lessons, but choosing which lessons to show for a new task is a credit-assignment problem. Capture-the-flag (CTF) gives a verifiable reward: the flag is checked by exact match. We ask whether a learner that sees only the flag, and which shown lessons appear in the agent's commands, can choose lessons better than a fixed relevance ranking. On a controlled synthetic benchmark it does, by 13 and 25 percentage points at two model sizes (95% CIs exclude zero). On a real CTF benchmark (NYU CTF Bench, 176 tasks, three seeds), static lessons added no detectable gain: $+0.6$ points $[-2.1, +3.4]$.**

***Index Terms*—LLM agents, agent memory, contextual bandits, verifiable reward, credit assignment, capture-the-flag**

I. Research Objectives

Agent memories that store lessons can encode wrong or useless ones [1], and agents tend to follow whatever past experience they are shown [2]. CTF removes a noisy judge of success from that choice; we test whether the flag alone is enough to learn which lessons help.

II. Methodology

A frozen LLM solves each task with shell commands in a container; three lessons are injected into its prompt. We compare no memory; a static top-3 ranked by a pretrained text-relevance model (a cross-encoder; on real CTFs, top-1 plus two diversity picks); and a learned contextual bandit whose lesson values are updated from the flag reward, crediting shown lessons whose command words appear in the agent's commands. Test beds: a synthetic decode microbenchmark (22 families, fresh random flags, Qwen2.5-14B/32B, held-out items), with all three arms; and NYU CTF Bench [3] (41 lessons written by the model from the files of 41 of 45 development tasks, 23 of them including a solution; scored on 176 disjoint test tasks, Qwen3.8-27B, three seeds, internet blocked), with no memory and static. Comparisons are paired by task with cluster-bootstrap 95% CIs.

III. Preliminary Results

On the microbenchmark, learned retrieval beats static by $+13.2$ $[+6.1, +20.9]$ points at 14B and $+25.2$ $[+16.5, +33.6]$ at 32B (Fig. 1); the same exploration without context gives no gain over static. Causal credit, which re-solves a task without each used lesson, adds $+10.6$ $[+5.4, +15.9]$ points over outcome credit at 32B (three draws, $\sim 6\times$ tokens) and no detectable gain at 14B. On NYU CTF Bench, pooled over 524 pairs, static lessons did not detectably change the held-out solve rate: 78 solved without memory, 81 with static lessons ($+0.6$ $[-2.1, +3.4]$; 25 pairs gained, 22 lost; sign test $p = 0.77$); on file-only tasks that need no live service the difference is $+1.7$ $[-4.0, +7.5]$ (a secondary analysis). A shown lesson appears in the agent's commands in 58 of 524 runs, against 46 by chance in the paired no-memory runs. Without memory the agent solves about 15% of runs, so the 45 development tasks would give a learner only about 7 solved runs per pass, against 510 to 1,245 in two passes on the microbenchmark.

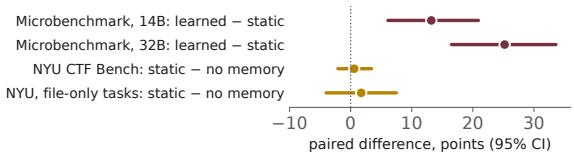


Fig. 1. Paired differences in solve rate, in points; right of 0 favors the first-named arm (95% bootstrap CIs, clustered by family on the microbenchmark, by task on NYU CTF Bench).

IV. Expected Contributions

An end-to-end pipeline that carries lesson memory from a controlled benchmark to a real CTF benchmark; a positive result for learned choice on the controlled benchmark; and, on real CTFs, a paired comparison that rules out gains from static lessons larger than about 3 points (upper CI bound $+3.4$), with candidate, untested explanations: few verified successes, a coarse keyword credit signal, and self-written lessons, 18 of 41 with no solution to learn from. The lesson pool is fixed; self-extending memory is not evaluated.

References

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[3] Shao *et al.*, "NYU CTF Bench: a scalable open-source benchmark dataset for evaluating LLMs in offensive security," NeurIPS Datasets and Benchmarks, 2024.